

B30801 Managerial Accounting Autumn 2025

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#1A: Cost Allocation and RegionFly Primer.

CHICAGO BOOTH



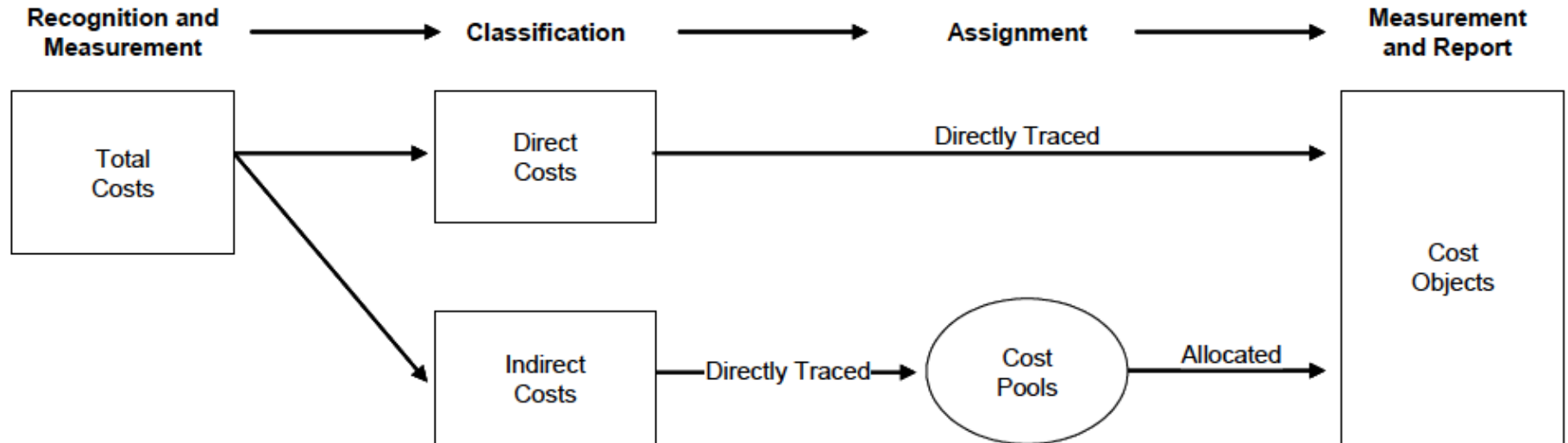
The University of Chicago Booth School of Business

Cost systems

Cost systems perform two functions:

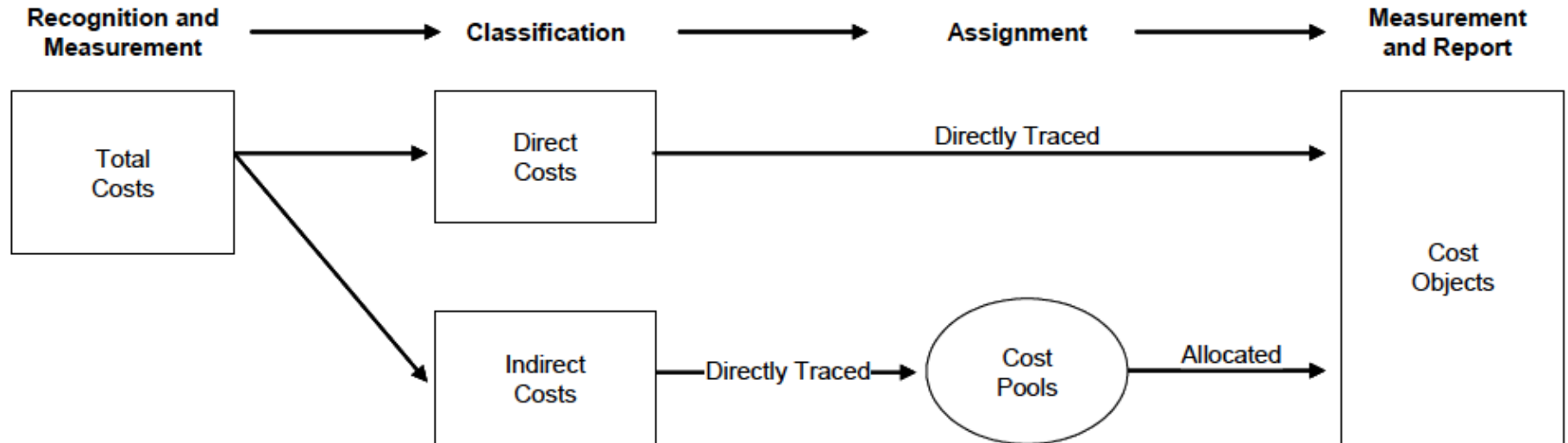
1. **Accumulate cost data:** as resources are acquired, the accounting system captures the associated costs (at some point, now or later, there is a cash outlay).
2. **Assign costs** to those activities, products, and services that create value (**cost objects**).
 - In RegionFly, the relevant cost objects are routes.
 - Done either **directly** (direct costs) or **indirectly** by way of an **allocation rule** (indirect costs).

Basic structure of Cost Systems.



Source: Introduction to Cost-Accounting Systems (HBS).

Basic structure of Cost Systems.



Whether and how we allocate costs depends on purpose, i.e., the decision at hand. There is no universally correct allocation rule.

Cost Allocation

Cost pools: buckets of accumulated costs we wish to assign to cost objects :

- For RegionFly, these are the **variable direct costs** (directly assigned to routes; see Ex. 1) and **overhead costs**, accumulated in different accounts (cost pools) according to type of cost (bottom of Ex. 1).
- The overhead costs must be allocated to routes in some way (they are indirect costs).

VARIABLE DIRECT COSTS	2015	2016	2017	2018
Route #				
Route #1	\$173,051	\$167,593	\$213,085	\$205,474
Route #2	\$170,589	\$195,632	\$0	\$0
Route #3	\$195,612	\$210,962	\$195,643	\$154,203
Route #4	\$104,711	\$107,596	\$0	\$0
Route #5	\$122,443	\$130,971	\$99,896	\$98,196
Route #6	\$78,872	\$79,338	\$91,589	\$84,332
Route #7	\$168,453	\$161,250	\$171,569	\$180,000
Total Variable Direct Costs	\$1,013,732	\$1,053,342	\$771,782	\$722,206

OVERHEAD

Account #				
1000	\$755,242	\$787,084	\$576,695	\$538,631
2000	\$134,865	\$138,900	\$117,983	\$111,485
3000	\$485,512	\$510,102	\$373,751	\$349,082
4000	\$755,242	\$783,004	\$761,080	\$717,969
5000	\$269,729	\$281,473	\$264,022	\$242,362
6000	\$53,946	\$56,287	\$53,585	\$47,592
7000	\$242,756	\$252,165	\$184,761	\$172,566
Total Overhead Costs	\$2,697,292	\$2,809,015	\$2,331,876	\$2,179,686

Exhibit 2 RegionFly Chart of Overhead Accounts

Account #	Account	Description
1000	Flight Support Expenses	Passenger service and flying operating expenses, personnel expenses and fringe benefits for engineers and flight attendants, pilots, co-pilots that are not assigned to a particular route
2000	Sales Expenses	Wages and fringe benefits of reservations personnel and sales agents. Other selling expenses, including ticketing, sales and promotions, company-wide advertising and publicity
3000	Maintenance Expenses	Wages and fringe benefits of repair and maintenance workers, related equipment, materials, oil, and repair expenses
4000	Fees and Taxes	Aircraft financing expenses (interests on capital leases or loans), insurance, operating lease payments, rental costs, taxes, and Federal Aviation Authority registration and airport fees (e.g., landing fees)
5000	Depreciation	Straight-line depreciation on owned aircrafts, maintenance facilities, buildings, and leased equipment (capital leases)
6000	Other Expenses	Administrative costs, including office supplies (i.e., books, charts) and legal costs
7000	Airport Expenses	Wages and fringe benefits of air traffic and ground service personnel, including gate agents, baggage handlers, baggage claim operators, and cargo operators

Cost Allocation

Cost allocation rule: the mechanism we use to assign costs in a given cost pool to the cost object. Also known as the **burden rate** or **allocation rate**.

→ For RegionFly, the company has chosen to use VDCs as the activity measure. For 2016:

$$\begin{aligned}\text{Rate} &= (\text{Total Costs in Pool})/\text{activity measure} \\ &= \$2,809,015/\$1,053,342 \\ &= \$2.667\end{aligned}$$

We allocate \$2.667 of overhead for every dollar of VDC incurred by a given route.

[illegible]

[illegible]

This shows the the allocation of costs to Route 1 (by applying the calculated overhead allocation rate of \$2.667 to VDCs for that route).

		2016 Overhead Cost Allocation/Gross Margins										
			Route 1	Route 2	Route 3	Route 4	Route 5	Route 6	Route 7	Total		
		Revenues	1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233		
		Variable Direct Costs	(167,593)	(195,632)	(210,962)	(107,596)	(130,971)	(79,338)	(161,250)	(1,053,342)		
		Allocated Overhead Costs	(446,931)	(521,704)	(562,586)	(286,933)	(349,269)	(211,576)	(430,016)	(2,809,015)		
		Profit Margin	434,273	112,678	380,271	129,024	327,886	182,318	284,426	1,850,876		
		GM %	41.41%	13.58%	32.96%	24.64%	40.57%	38.53%	32.48%	32.40%		
		Total Overhead	2,809,015									
		Total VDC	1,053,342									
		Allocation rate =	2.6667645									

Now includes allocated overhead costs for the remaining routes.

[illegible]

With all costs now allocated, we can compute the profit margins, the focus of the RegionFly management team.

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The essence of the case is to evaluate whether using this (profit margin %) to decide what routes (products) to drop makes sense.